Test Plan Document

Traffic Management System

9 Test Cases — Unit, Integration and System Testing

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
UT-01	Authentication Module	To test successful user login with valid credentials.	Registered user account and accessible login page.	1. Open Login. 2. Enter username and password. 3. Click Login.	Username: valid_user Password: Valid@123	User is authenticated and the dashboard is displayed.	Not executed — manual testing required.	Not Run
UT-02	Authentication Module	To test login validation with invalid credentials.	Login page is accessible; registered account exists.	1. Open Login. 2. Enter invalid username/password. 3. Click Login.	Username: invalid_user Password: wrong123	Login is rejected and a clear authentication error is displayed.	Not executed — manual testing required.	Not Run
UT-03	Incident Reporting Module	To test submission of a valid traffic incident report.	User is authenticated and incident-reporting page is available.	1. Open Incident Reporting. 2. Select incident type. 3. Enter location and description. 4. Submit.	Type: Accident Location: MG Road Description: Two vehicles blocking lane	Input is accepted, the incident is stored, and a success/status message is displayed.	Not executed — manual testing required.	Not Run
IT-01	Traffic Data & Live Map	To test integration between traffic data and the live map.	User is logged in; traffic data and map component are available.	1. Open Live Traffic Map. 2. Load traffic data. 3. Observe map indicators.	Road: MG Road Traffic level: High Incident: Accident	Traffic data is retrieved and correctly represented on the map with congestion/incident indicators.	Not executed — manual testing required.	Not Run
IT-02	Route Planning	To test integration of route planning with traffic data.	User is logged in; source/destination and traffic data are available.	1. Enter source and destination. 2. Request route. 3. Compare route with traffic information.	Source: Majestic Destination: Whitefield Traffic: High on Route A	System considers available congestion data and displays a suitable/least-congested route.	Not executed — manual testing required.	Not Run
IT-03	Alerts & Notifications	To test integration of traffic events with user alerts.	Notification mechanism is available and relevant user is registered.	1. Create/load a traffic incident. 2. Trigger alert generation. 3. Check user notifications.	Event: Road closure Area: MG Road	Relevant users receive/display a traffic alert about the road closure.	Not executed — manual testing required.	Not Run
ST-01	Traffic Management System	To test end-to-end traffic monitoring and route planning.	System is deployed; user is authenticated; map and traffic data are available.	1. Login. 2. Open live map. 3. Enter source/destination. 4. Request route. 5. View alerts.	Source: Majestic Destination: Whitefield Incident: Accident	System successfully authenticates the user, shows traffic/map data, calculates a suitable route, and displays relevant alerts.	Not executed — manual testing required.	Not Run
ST-02	Traffic Administration	To test administrator signal management with authorization and audit logging.	Authorized administrator is authenticated; signal data exists.	1. Open Signal Management. 2. Select a signal. 3. Change status. 4. Save. 5. Check audit log.	Signal: Junction-01 Old: RED New: GREEN	Authorized change is validated, stored, reflected in status, and recorded in the audit log.	Not executed — manual testing required.	Not Run
ST-03	Emergency Vehicle Priority	To test end-to-end emergency priority and optimized routing.	Verified emergency operator is authenticated; traffic data is available.	1. Request emergency priority. 2. Enter/select destination. 3. Request optimized route. 4. Verify request log.	Vehicle: Ambulance-01 Destination: City Hospital Traffic: High	Authorized emergency request is accepted, an optimized route is provided, and the priority action is logged.	Not executed — manual testing required.	Not Run